SMARTBRIDGE
Let's Bridge the Gap

SKILL WALLET
A SmartBridge Product

## Scalability & Future Plan

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| **Date** | 03 JULY 2026 |
| **Team ID** | XXXXX |
| **Project Name** | **A Comprehensive Measure of Well-Being** |
| **Maximum marks** | 1 Mark |

**Scalability & Future Plan:**
This document outlines how the current project solution can be scaled to handle larger user bases, increased data loads, or extended features in the future. It also captures the team's roadmap for enhancing and evolving the project beyond its current state.

**Current System Limitations:**

| S.No | Limitation | Impact | Priority to Address |
|---|---|---|---|
| 1 | The application currently supports a limited number of concurrent users. | Performance may decrease when many users access the system simultaneously. | **High** |
| 2 | The model is trained using a static HDI dataset. | Predictions may not reflect the latest global development statistics. | **Medium** |
| 3 | The application is deployed on a local Flask server. | Limits accessibility and scalability for public users. | **High** |

**Scalability Plan**

| S.No | Scalability Aspect | Current State | Proposed Upgrade / Solution |
|---|---|---|---|
| 1 | **User Load** | Supports only local users during testing. | Deploy the application on cloud platforms (Render, AWS, Azure, or Google Cloud) with load balancing to support thousands of users. |
| 2 | **Data Storage** | Uses a CSV file as the primary dataset. | Migrate to a relational database such as MySQL or PostgreSQL for efficient data management and scalability. |
| 3 | **Performance** | Machine learning model executes on a single server. | Optimize the model, use caching, and improve backend performance for faster response times. |
| 4 | **Security** | Basic input validation is implemented. | Add user authentication, HTTPS encryption, secure APIs, and role-based access control for enhanced security. |

**Future Roadmap**

| Phase | Planned Feature / Enhancement | Target Timeline | Expected Impact |
|---|---|---|---|
| Phase 1 | Deploy the HDI Insight application to a cloud platform with a public URL. | Next 1 Month | Improve accessibility and allow users to access the application from anywhere. |
| Phase 2 | Integrate real-time HDI data updates from trusted international sources. | Next 3 Months | Ensure predictions are based on the latest development statistics. |
| Phase 3 | Introduce advanced machine learning algorithms and comparative country analysis. | Next 6 Months | Increase prediction accuracy and provide deeper analytical insights. |
| Phase 4 | Develop a mobile application with interactive dashboards and multilingual support. | Next 12 Months | Enhance user experience, accessibility, and reach a broader global audience. |